European Commission

Defence Industry and Space

Cyber in Defence

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DEFENCE APPLICATIONS



Cyberspace has become an increasingly contested domain and a field for strategic competition



Cyber-attacks, espionage and disinformation campaigns targeting the EU and its Member States, including defence actors, are on the rise



Significant cybersecurity incidents can disrupt or damage critical infrastructure that our armed forces rely on

Why does cyber matter for defence?

Generally, the cyber domain refers to the virtual environment created by interconnected information systems, networks, and digital infrastructures. In the defence context, it encompasses all activities related to **securing, disrupting, or exploiting digital networks and data**.

Key defence use cases include situational awareness; cyber threat detection and analysis; secure communications; encryption and data protection; critical infrastructure protection; incident response; offensive cyber operations; and cyber education and training.

In 2018, the EU formally recognised cyberspace as the **fifth domain of military operations**, alongside land, sea, air and space. Cyberspace is a crosscutting and critical enabler that underpins and connects all others.

In the coming years, **several disruptive technologies will play a key role in shaping the future of cyber defence** by transforming military processes and capabilities. These include artificial intelligence (AI), blockchain technology, the Internet of Things (IoT), the expansion of next-generation networks (such as 5G and beyond), cloud computing, and quantum computing.

What does the EU aim to achieve?

The priorities outlined in the White Paper on European Defence - Readiness 2030 aim at achieving **full-spectrum cyber defence capabilities**. For the first time at EU level, the White Paper makes the case to develop together with Member States a voluntary support scheme for offensive cyber capabilities as credible deterrence.

Key actions under the European Defence Fund (EDF) to strengthen EU leadership in cyber defence include **enhancing situational awareness**, investing in skills development via **cyber ranges**, building **secure military cloud** and communications networks, and deploying **Cyber Rapid Response Teams** with advanced tools and training.

EDF priorities address key capability gaps to enhance EU's resilience and strategic autonomy in the cyber domain. Moreover, EDF actions also contribute to achieving the **EU Cyber Defence Policy** objectives.

Main advantages of cyber applications in defence

- Real-time situational awareness
- Faster threat detection
- Rapid response and flexibility
- Non-lethal deterrence
- Limited physical and human damage
- Tactical advantage when used in combined arms

Main vulnerabilities arising from cyber applications in defence

- Broken encryption standards
- Affects essential infrastructure
- Espionage
- Compound risks with Artificial Intelligence
- Data manipulation and theft
- Attribution challenges and accountability

When will EDF project results be translated into military capabilities?

A total of **€200 million from the EDF** has been dedicated to a range of cyber-related research and development actions.

A successful example is the CYBER4DE project, funded through the European Defence Industrial Development Programme (EDIDP), which has developed a toolbox to detect, investigate and remedy hostile cyber actions. CYBER4DE has already delivered concrete capabilities which are used in an operational context**. The EU Cyber Rapid Response Team (led by Lithuania), using the toolbox, assisted Ukraine at the beginning of the war** and helped the country's institutions to face cybersecurity threats. The toolbox has also been used in deployments to Mozambique and Republic of Moldova under a CSDP mandate.

Other notable examples include the projects ECYSAP and PANDORA. ECYSAP was planned to be used by the EUMS and MPCC for EU missions and operations, and now has a follow-on EDF project (ECYSAP-EYE), aiming to develop a European platform for real-time cyber situational awareness with rapid response capabilities and decision-making support for military end-users. **PANDORA** focused on implementing a technical solution for real-time threat hunting and incident response, being used by several MODs and within the CTIRISP PESCO project, alongside results from the follow-on EDF project **AInception**.

Projects from precursor programme EDIDP currently delivering results

CYBER4DE
Cyber Rapid Response Toolbox for Defence Use
EU financing: €9.3M

ECYSAP
European Cyber Situational Awareness Platform
EU financing: €10.9M

PANDORA
Cyber Defence Platform for Real-time Threat Hunting, Incident Response and Information Sharing
EU financing: €6.8M

EDF projects expected to deliver results in 2025-2027

EUCINF
EUropean Cyber and Information warfare toolbox
EU financing: €32.9M

ACTING
Improved capacity for cyber training and exercises
EU financing: €16.2M

EU-GUARDIAN
European framework and proofs-of-concept for the intelliGent aUtomAtion of cyber Defence Incident mAnageMeNt
EU financing: €13.5M

FACT
Federated Advanced Cyber physical Test range
EU financing: €26.9M

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